

Financial Data Analytics Group Project

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Introduction

Derivative instruments have become an increasingly important part of corporate risk management over the past several decades. Non-financial firms use derivatives such as interest rate, currency, and commodity contracts to reduce exposure to risks from fluctuations in market conditions that can significantly impact earnings, cash flows, and firm value. Over the last two decades, firms have faced increased exposure to exchange rate volatility, commodity price swings, and interest rate risk, particularly during periods of economic turmoil such as the 2008 financial crisis and the COVID-19 pandemic. As businesses expand into global markets and these exposures have larger financial effects, firms have faced growing pressure from shareholders and stakeholders to hedge the risks and enhance firm value without negatively impacting profitability. In response, many corporations have incorporated derivative strategies into their broader financial and operational risk management practices.

Derivative instruments derive their value from an underlying asset, rate, or market condition and are commonly used by firms to hedge financial exposures. Interest rate derivatives are often used to manage borrowing costs and protect against fluctuations in market interest rates that impact debt offerings and bond portfolios. Currency derivatives help firms reduce the risks of foreign exchange rate movements tied to international revenues, expenses, and investments. Small changes in currency conversion rates can have large impacts on multinational firms' reporting and valuations. Commodity derivatives are used to stabilize costs and reduce exposure to price volatility in raw materials such as oil and natural gas, metals, and agricultural products. While derivatives can also be used for speculative purposes, non-financial corporations primarily utilize these tools as part of broader risk management strategies and cash flow stabilization.

This report analyzes derivative usage among non-financial S&P 1500 firms from 2000 to 2024 using a panel dataset compiled from company 10-K annual report filings. Firms are required to disclose their material risks and the quantitative and qualitative measures they use to address them, including their derivative usage. The analysis focuses on trends in firm derivative usage over time, the differences in usage across various industries, and factors associated with the adoption of derivative instruments. In particular, the report examines the use of interest rate, currency, and commodity derivatives and evaluates how usage patterns evolved through major economic events including the 2008 financial crisis and the COVID-19 pandemic period.

Multiple data analytics techniques are used throughout the analysis, including time-series analysis, cross-sectional comparisons, industry-level analysis using 2-digit SIC classifications, and regression analysis. The findings provide insights into how firms manage their financial risk exposures and how derivative usage has changed over time as firms have become increasingly exposed to global financial and operational risks.

Data Description

This section describes the structure and characteristics of the derivatives disclosure dataset used in the empirical analysis. The analysis focuses on firm-level interest rate, currency, and commodity derivative exposures across multiple sample years. Summary statistics, and exposure trends are presented to better understand the distribution and evolution of corporate derivatives usage over time

The combined dataset contains 17,430 firm-year observations across 2,456 unique firms identified by GVKEY. The sample covers seven fiscal years: 2000, 2004, 2008, 2012, 2016, 2020, and 2024. Since most firms appear in all seven sample years, the dataset is close to a balanced panel. However, because some firms appear in fewer years, the final structure is an unbalanced firm-year panel.

Table 1: Yearly Derivative Usage and Aggregate Notional Exposure by Derivative Type

Year	IR Users	IR Total Notional (\$bn)	Currency Users	Currency Total Notional (\$bn)	Commodity Users	Commodity Total Notional (\$bn)
2000	313	347.3	278	248.6	56	318.5
2004	506	370.4	420	176.0	100	59.9
2008	361	247.8	356	231.3	42	9.0
2012	246	349.5	352	459.4	51	37.3
2016	292	453.9	380	621.0	54	61.3
2020	334	1,514.2	378	1,112.5	84	138.9
2024	216	810.4	273	1,053.5	60	177.5

User counts include only observations where the Derivative User indicator equals 1, while observations with 0 or -1 values in the Notional Amount of Interest Rate Derivatives field were excluded from the analysis, and the sum of the remaining valid notional amounts was considered. This is followed across the whole analysis.

Table 1 presents yearly derivative usage counts and aggregate notional exposure for interest rate, currency, and commodity derivatives from 2000 to 2024. The table shows that derivative usage and exposure vary meaningfully across both time and derivative type. Interest rate derivatives have the highest aggregate notional exposure in most years, reaching a peak of \$1,514.2 billion in 2020. Currency derivative exposure rises sharply after 2012, increasing from \$459.4 billion in 2012 to \$1,112.5 billion in 2020 and \$1,053.5 billion in 2024. Commodity derivative usage remains lower than interest rate and currency derivative usage throughout the sample, although commodity notional exposure increases after 2016. Overall, the table suggests that derivative exposure is not explained only by the number of firms using derivatives. Instead, total exposure appears highly concentrated among a smaller group of firms with large notional positions.

The sharp increase in derivative notional exposure in 2020 may reflect heightened hedging activity during the COVID-19 period and historically volatile interest rate conditions.

Figure 1: Interest Rate, Currency, and Commodity Users Over Time

Figure 1 illustrates the number of firms using interest rate, currency, and commodity derivatives across the sample period. Interest rate and currency derivatives consistently exhibit substantially higher adoption levels than commodity derivatives throughout the dataset. Interest rate derivative usage rises sharply between 2000 and 2004 before gradually declining in subsequent years. In contrast, currency derivative usage remains relatively stable after 2008 and eventually exceeds interest rate derivative usage after 2012. Commodity derivative usage remains comparatively limited across all years, indicating that commodity hedging is concentrated within a narrower

subset of firms and industries. The temporary increase in derivative usage during 2020 may reflect heightened market uncertainty and increased corporate hedging activity during the COVID-19 period.

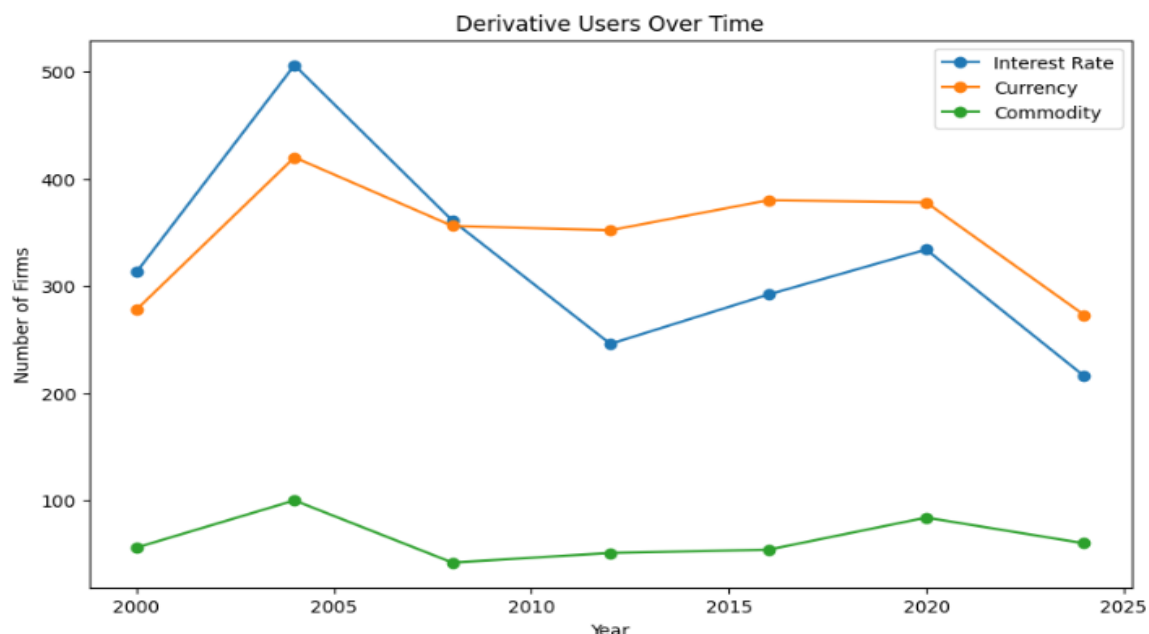


Figure 2: Aggregate Notional Exposure Over Time

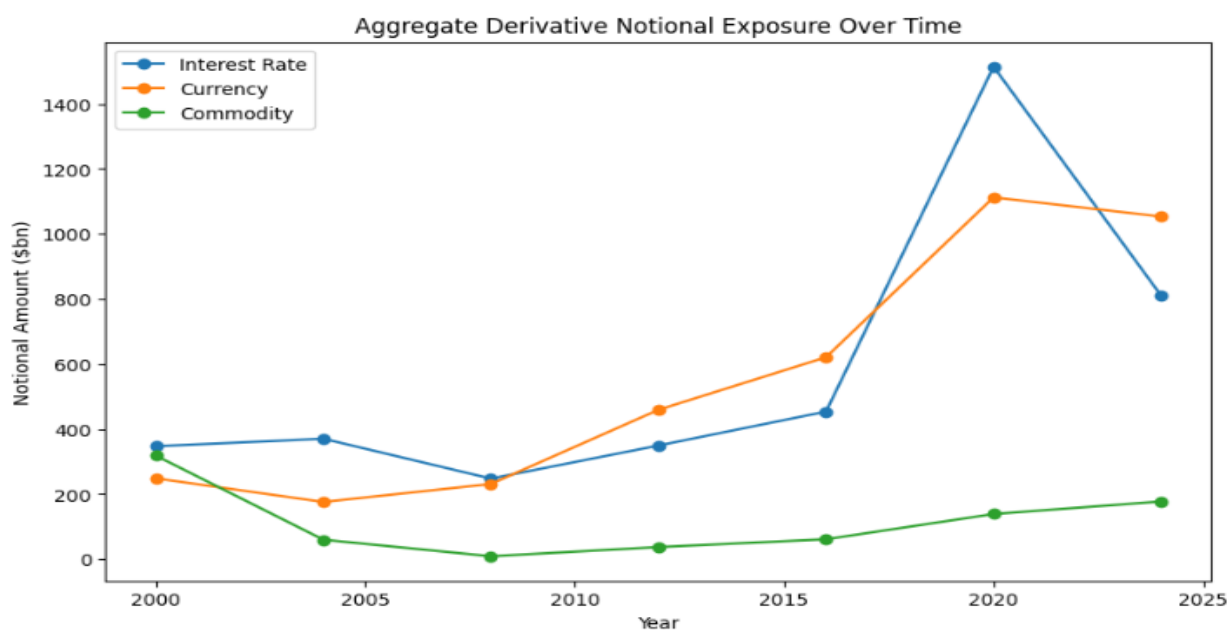


Figure 2 presents aggregate derivative notional exposure across interest rate, currency, and commodity derivatives over time. Interest rate derivatives exhibit the largest notional exposures throughout most of the sample period, with exposure increasing sharply to over \$1.5 trillion in 2020 before declining in 2024. Currency derivative exposure also rises substantially after 2012 and continues increasing through 2024, eventually exceeding interest rate derivative exposure in the final sample year. In contrast, commodity derivative exposure remains comparatively small throughout the sample, although it gradually increases after 2012. The figure highlights that aggregate derivative exposure does not move proportionally with the number of derivative users. Instead, total exposure appears highly concentrated among a relatively small number of firms maintaining very large derivative

positions. The sharp increase in exposure during 2020 may reflect heightened hedging demand during the COVID-19 period and elevated financial market uncertainty.

Figure 3: Correlation Heatmap of Derivative Notional Exposures

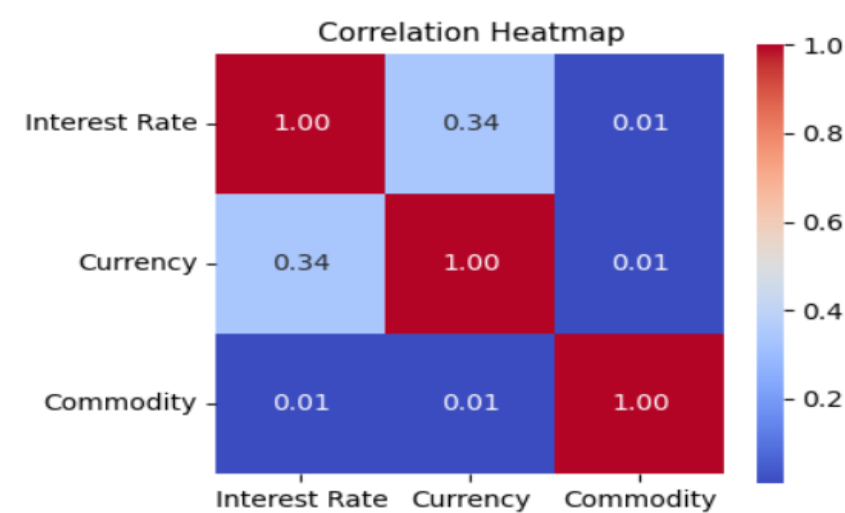


Figure 3 shows the correlation heatmap for notional exposures across the three derivative types. Interest rate and currency exposures show a moderate positive correlation of 0.34, suggesting that firms managing interest rate risk tend to also hedge currency risk, which is consistent with the behavior of large multinationals with more complex risk management needs. Commodity derivative exposure, on the other hand, shows almost no correlation with either, pointing to more specialized, industry-specific drivers rather than broad financial risk management.

Overall, corporate derivative activity appears highly concentrated among a subset of firms with large notional positions, with interest rate and currency derivatives dominating and commodity usage remaining limited and industry-specific. The skewness and concentration in the data motivate the use of further empirical analysis, including trend analysis, regression modeling, clustering, and scenario-based risk assessment.

Interest Rate Derivative Analysis

Industry Analysis

Table 2: Top Industries by Interest Rate Derivative Notional Exposure

SIC 2-Digit	Industry	IR Total Notional (\$bn)	Number of Firms
37	Transportation Equipment	1,190.4	63
73	Business Services	418.4	303
48	Communications	298.7	69
28	Chemicals	289.7	213
38	Scientific Instruments	277.4	159
49	Utilities	274.4	136
35	Industrial Machinery	143.4	165
36	Electrical Equipment	124.4	199
99	Nonclassifiable Establishments	122.1	9
20	Food Products	100.9	68

Table 2 reports the top industries by aggregate interest rate derivative notional exposure. Transportation Equipment dominates the sample with \$1,190.4 billion in aggregate IR notional exposure across 63 firms, indicating substantial interest rate risk management activity in this sector. Business Services, Communications, Chemicals, Scientific Instruments, and Utilities also report large notional exposures, suggesting that interest rate hedging is concentrated among capital-intensive and large-scale operating sectors. The results show that IR derivative exposure is not evenly distributed across industries but is instead concentrated in a small group of sectors with larger financing needs and more complex treasury operations.

Figure 4: Top Industries by Interest Rate Derivative Exposure

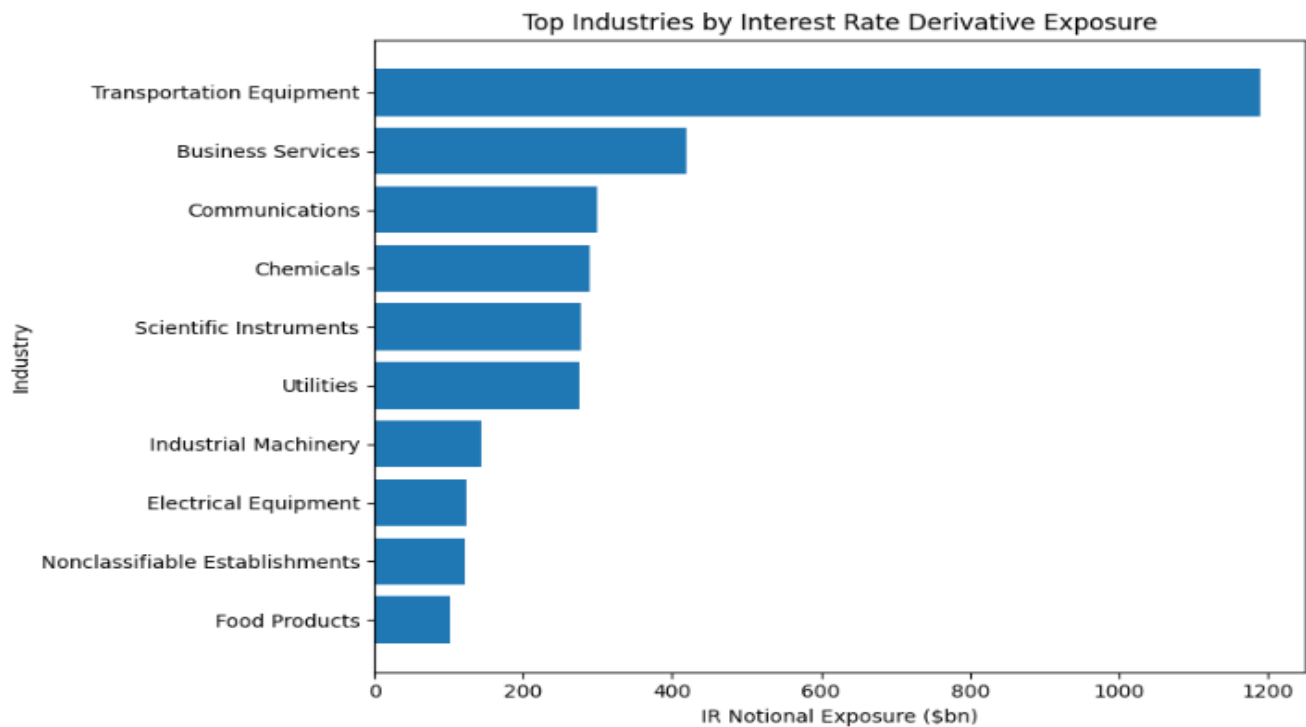


Figure 4 presents aggregate interest rate derivative exposure by industry across the full sample period from 2000 to 2024. The chart aggregates total notional exposure across all firms and sample years within each industry. Transportation Equipment exhibits substantially higher cumulative exposure than all other industries, followed by Business Services, Communications, Chemicals, Scientific Instruments, and Utilities. The concentration of exposure among these industries suggests that long-term interest rate hedging activity is primarily associated with sectors characterized by large financing requirements, capital-intensive operations, and more sophisticated treasury management practices. The results further indicate that derivative exposure is highly concentrated among a relatively small number of industries over the sample period

Regression Analysis

Table 3: Regression Analysis of Interest Rate Derivative Exposure

Variable	Coefficient	t-Statistic	p-value
Intercept	-114.63	-11.535	0.000
Year	0.0598	12.090	0.000

Table 3 presents the regression results examining the relationship between time and interest rate derivative exposure. The dependent variable is the logarithm of positive interest rate derivative notional amounts. The

coefficient on the Year variable is positive and statistically significant at the 1% level, indicating that interest rate derivative exposure has increased systematically over time. The estimated coefficient suggests that IR derivative exposure increases by approximately 5.96% per sample year. Although the model exhibits a relatively low R-squared value, this is expected given the substantial firm-level heterogeneity in derivative exposure across industries and firms. Overall, the regression results support the conclusion that firms engaging in interest rate hedging have maintained increasingly larger derivative positions over the sample period.

Clustering Analysis

Figure 5: Clustering of Interest Rate Derivative Users

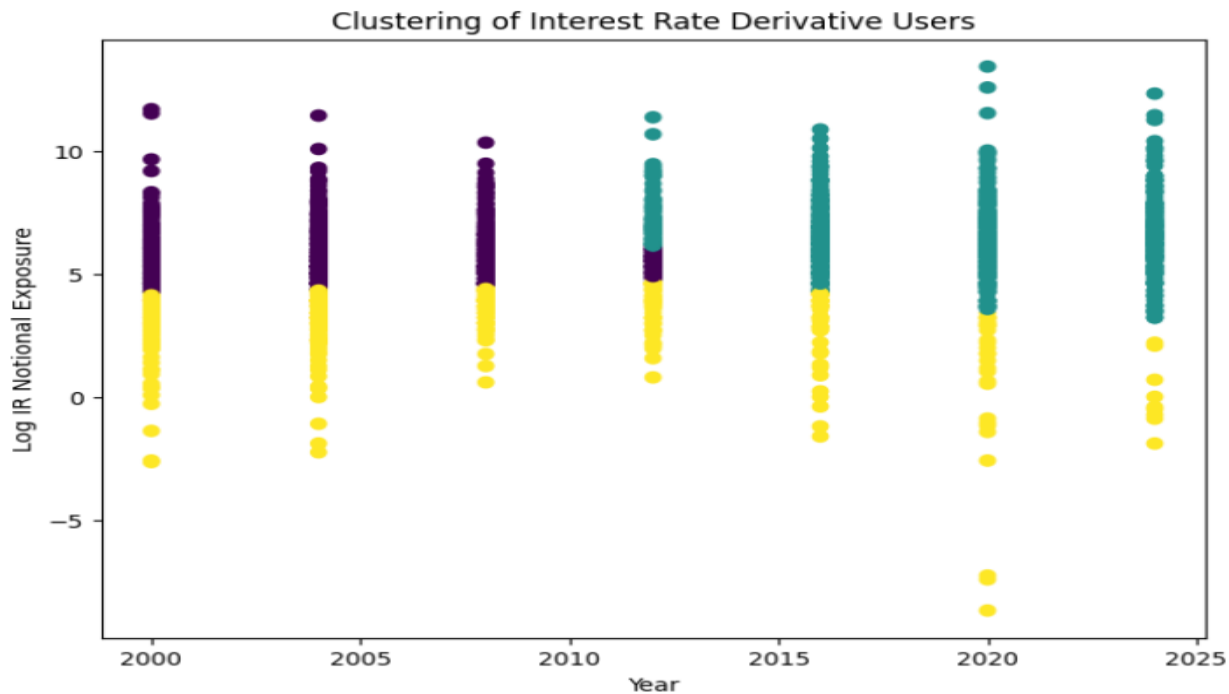


Figure 5 presents the clustering analysis of firms using interest rate derivatives based on log-transformed IR notional exposure and time. The clustering results identify three distinct groups of firms characterized by different hedging intensity levels. The first cluster consists of firms with relatively small derivative exposures, while the second cluster contains firms with moderate and more stable hedging positions. The third cluster captures firms with substantially larger derivative exposures and represents the most intensive users of interest rate derivatives. The visualization further indicates that higher exposure clusters become more prominent in later sample years, particularly after 2012 and during the 2020 period. Overall, the clustering analysis suggests that corporate interest rate hedging behavior is highly heterogeneous and concentrated among a subset of firms with significantly larger derivative positions.

Figure 6: Distribution of Log Interest Rate Derivative Notional Exposure

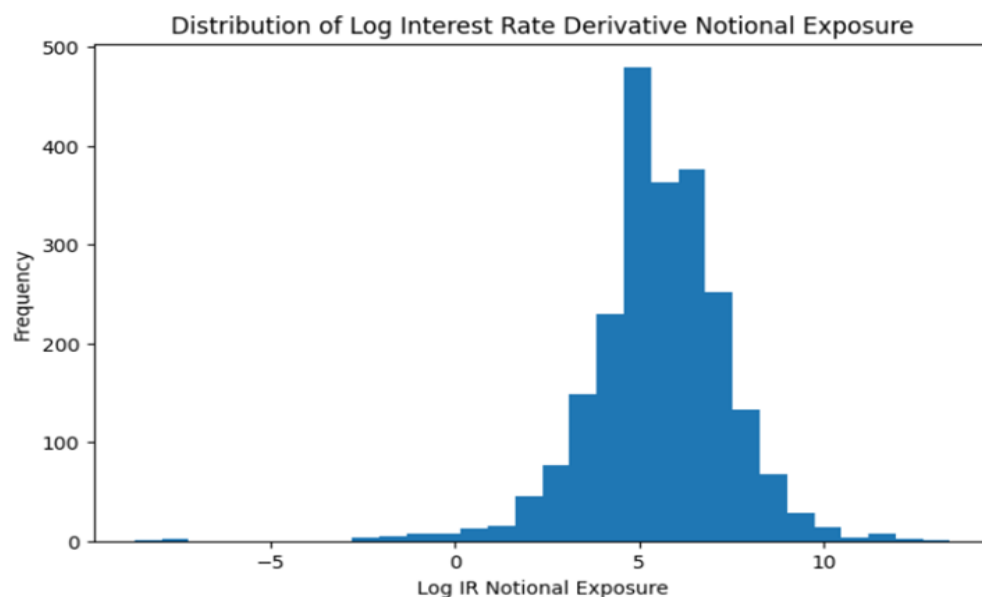


Figure 6 shows the distribution of log-transformed interest rate derivative notional exposure for firms with positive derivative positions. The raw data is heavily skewed, with a handful of firms carrying much larger positions than the rest, so a log transformation was applied to normalize it. Even so, a moderate right tail remains. Most firms cluster around mid-range exposure levels, and the overall pattern suggests that derivative activity tends to be concentrated among a smaller subset of firms running large hedging programs.

Sensitivity Analysis

Table 4: Industry Sensitivity to a 1% Interest Rate Shock

Industry	IR Total Notional (\$bn)	Number of Firms	Estimated Exposure Under 1% Shock (\$bn)
Transportation Equipment	1,190.4	63	11.90
Business Services	418.4	303	4.18
Communications	298.7	69	2.99
Chemicals	289.5	213	2.90
Scientific Instruments	277.4	159	2.77
Utilities	274.4	136	2.74
Industrial Machinery	143.4	165	1.43
Electrical Equipment	124.4	199	1.24
Nonclassifiable Establishments	122.1	9	1.22
Food Products	100.9	68	1.01

Table 4 presents an illustrative sensitivity analysis based on a hypothetical 1% interest rate shock applied to aggregate industry-level interest rate derivative exposure. The analysis estimates relative exposure sensitivity by applying a proportional 1% shock to cumulative notional exposure across industries. Transportation Equipment exhibits the highest estimated sensitivity, with approximately \$11.9 billion of exposure under the assumed shock scenario. Business Services, Communications, Chemicals, Scientific Instruments, and Utilities also display relatively large estimated sensitivities due to their substantial cumulative derivative positions. The results indicate that industries with greater financing requirements, higher leverage, and more complex treasury operations

maintain significantly larger interest rate hedging exposures. The findings further suggest that interest rate risk management activity is highly concentrated among a relatively small subset of industries and firms. While the analysis represents a simplified exposure-based approximation rather than a full valuation model, it provides useful insight into the relative concentration of interest rate sensitivity across industries.

Currency Derivative Analysis

Time Series Analysis

Figure 7: Currency Derivative Users and Aggregate Notional Exposure Over Time

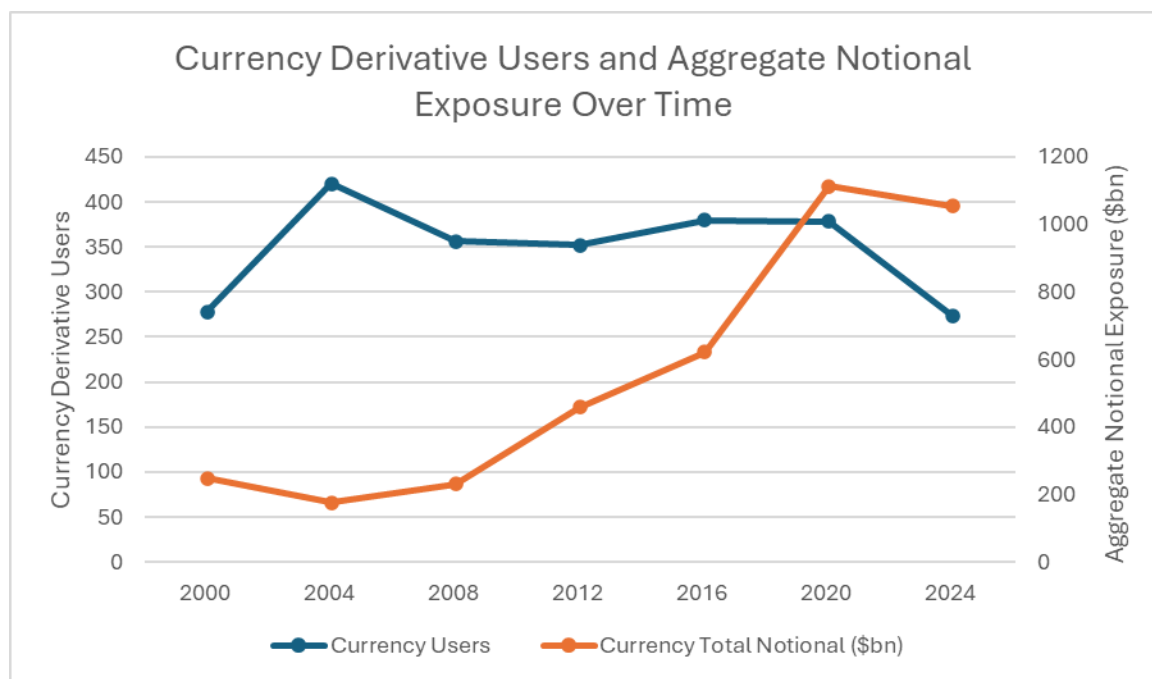


Figure 7 summarizes currency derivative usage among non-financial S&P 1500 firms from 200 to 2024. The number of firms reporting currency derivative usage increased from 278 firms in 2000 to a peak of 420 firms in 2004. Following this increase, usage remained relatively stable throughout the rest of the sample period, fluctuating between approximately 350 and 380 firms from 2008 through 2020, before declining to 273 firms in 2024. The persistence of several hundred currency derivative users throughout the sample suggests that foreign exchange rate management remained an important component of corporate risk management practices. The relatively stable participation rates after 2004 indicate that the adoption of currency derivatives became well established among firms with significant international operations and exposure to foreign exchange risk.

While the number of firms using currency derivatives remained relatively stable, aggregate currency derivative notional exposure increased significantly over the sample period. Total reported currency derivative exposure increased from approximately \$248.6 billion in 2000 to more than \$1 trillion by 2020 and 2024. Exposure remained relatively moderate during 2000-2008 as firms began adopting the use of currency derivatives. Following the 2008 financial crisis, aggregate exposure increased from \$231.2 billion in 2008 to \$459.4 billion in 2012, nearly doubling the total notional exposure. This growth may reflect heightened attention to risk management practices, increased international operations, and greater use of foreign exchange hedging among multinational firms. Additional growth occurred after 2012, with aggregate notional exposures reaching \$621.0 billion in 2016 before exceeding \$1 trillion by 2020. The largest increase occurred during the COVID-19 period, when disruptions to global supply chains, international trade, and foreign exchange markets created elevated

uncertainty and volatility for multinational firms. Although exposure declined slightly from its 2020 peak, aggregate currency derivative positions remained historically high in 2024, indicating that firms continued to actively manage substantial foreign exchange risk.

The divergence between relatively stable user counts and rapidly increasing notional exposure provides additional insight into how firms manage foreign exchange risk. Although the number of firms utilizing currency derivatives changed relatively little over the sample period, aggregate exposure increased more than fourfold. This suggests that existing users were managing increasingly large foreign exchange positions rather than that currency derivative adoption was becoming substantially more widespread. As firms expanded internationally and became more integrated into global markets, their exposure to exchange rate fluctuations increased. The results indicate that currency risk management became more intensive among existing users, reflecting the growing importance of foreign exchange hedging in managing multinational operations and cross-border business activities.

Industry Analysis

Table 5: Top Industries by Currency Derivative Notional Exposure

SIC 2-Digit	Industry	Currency Total Notional (\$bn)	Number of Firm Year Observations
28	Chemicals	659.5	299
35	Industrial Machinery	642.5	326
73	Business Services	560.0	354
37	Transportation Equipment	402.3	145
36	Electrical Equipment	293.6	309
38	Scientific Instruments	259.9	280
48	Communications	179.7	56
20	Food Products	158.4	120
59	Misc. Retail	104.3	32
21	Tobacco Products	92.7	8

Figure 8: Top Industries by Currency Derivative Exposure

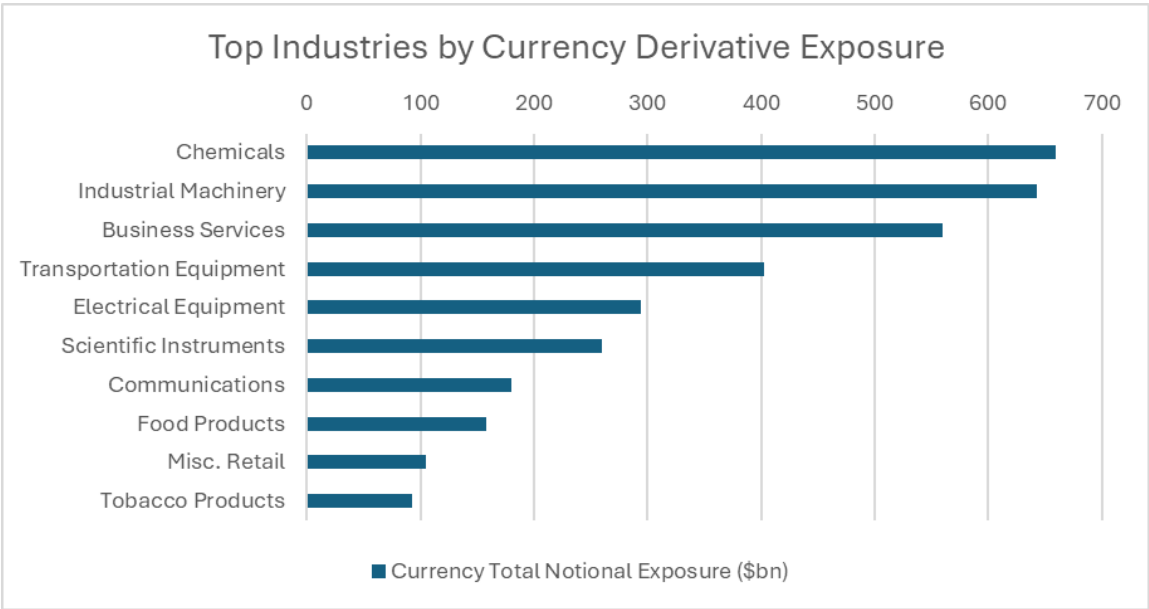


Table 5 and Figure 8 report on the industries with the highest aggregate currency derivative notional exposure across the full sample period. Chemicals, Industrial Machinery, and Business Services report the largest cumulative currency derivative positions, followed by Transportation Equipment, Electrical Equipment, and Scientific Instruments. Many of the industries with the highest exposures are globally integrated manufacturing and production sectors that rely on international supply chains, foreign subsidiaries, imported material inputs, and export markets. These firms face significant exchange rate risk through both their operations and revenues, creating a strong incentive to hedge foreign currency exposures.

An interesting exception is the presence of Business Services and Communications among the top industries by currency derivative exposure. Unlike manufacturing-focused sectors, these industries are less dependent on physical supply chains and are more likely to be exposed to currency risk through international operations, foreign customers, and revenues earned in multiple currencies. While aggregate exposure is generally highest in industries with large numbers of firm-year observations, Communications represents an exception, reporting nearly \$180 billion in aggregate currency derivative exposure with only 56 firm-year observations. This suggests that foreign exchange risk may be concentrated among a relatively small number of large multinational firms within the industry. Overall, the results suggest that currency derivative usage is associated not only with global production networks but also with firms maintaining broad international business footprints and significant economic activity across borders.

Commodity Derivative Analysis

Time Series Analysis

Figure 9: Commodity Derivative Usage

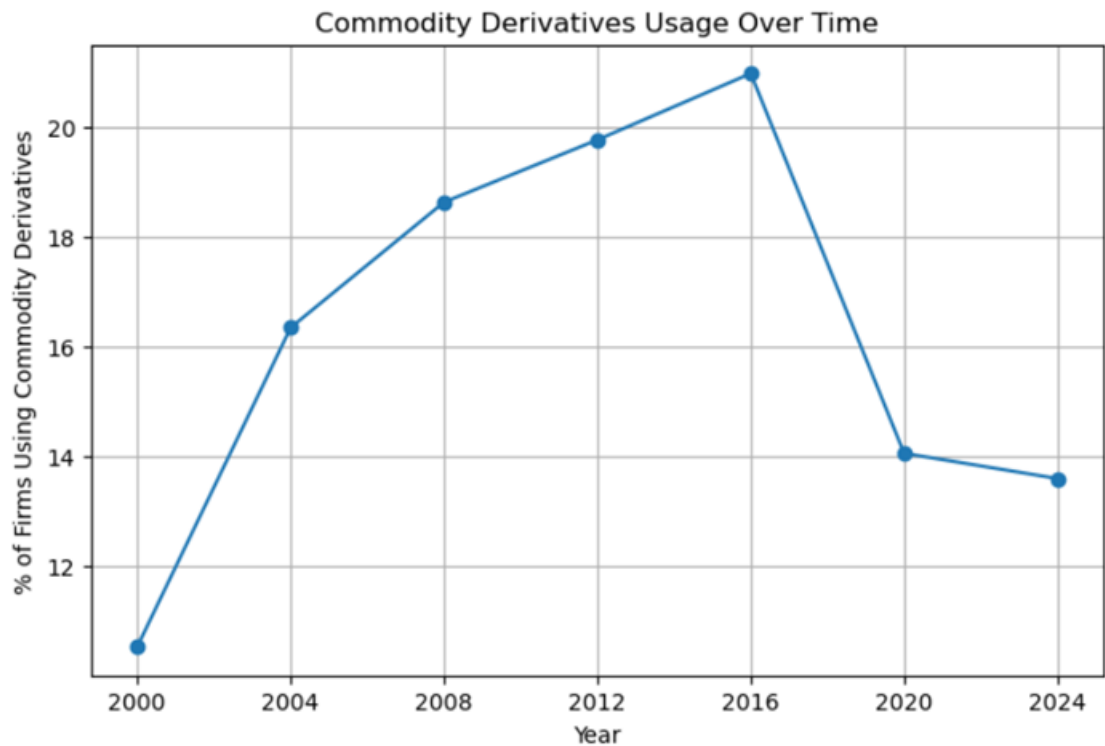


Figure 9 presents the percentage of active firms using commodity derivatives from 2000 through 2024. Commodity derivative usage increased steadily from approximately 10.5% of firms in 2000 to a peak of roughly 21.0% in 2016. Following 2016, usage rates declined to approximately 13–14% by 2020 and 2024.

The upward trend through the mid-2010s may reflect increased adoption of commodity risk management practices during periods of commodity price volatility and global economic expansion. Firms with significant exposure to energy, raw materials, or agricultural inputs may have increasingly utilized derivatives to hedge against fluctuations in input costs and market uncertainty.

The decline in usage observed after 2016 may indicate changes in corporate hedging strategies, shifts in industry composition, or reduced commodity exposure among firms remaining in the sample. Despite the decline in the percentage of users, commodity derivatives continued to play an important role in risk management for firms exposed to commodity price risk.

Cross-Sectional Analysis

Figure 10: Distribution of Commodity Derivative Notional Amounts

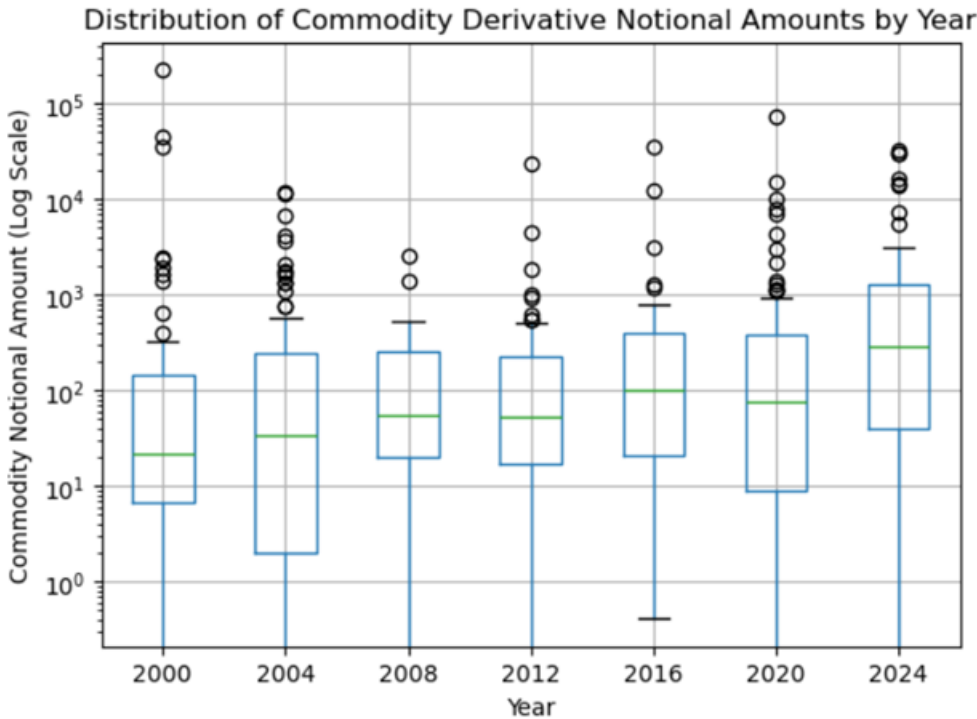


Figure 10 shows the distribution of commodity derivative notional amounts among firms using commodity derivatives across the sample period. The distributions are highly right-skewed in every year, with most firms reporting relatively modest notional amounts and a small number of firms holding substantially larger derivative positions. This pattern is reflected in the numerous outliers and the wide range of notional values shown in the boxplots.

Median notional amounts generally increased over time, particularly in 2016, 2020, and 2024. This suggests that although the percentage of firms using commodity derivatives declined somewhat in recent years, users of derivatives maintained larger positions in later years. The increasing spread of the distributions also indicates greater variation in commodity derivative exposure across firms over time.

The large difference between mean and median notional amounts further highlights the uneven distribution of commodity derivative activity. This pattern suggests that firms facing greater exposure to commodity price fluctuations tend to maintain larger hedging positions than firms with more limited commodity-related risks.

Industry Analysis

Figure 11: Commodity Derivative Use by Industry

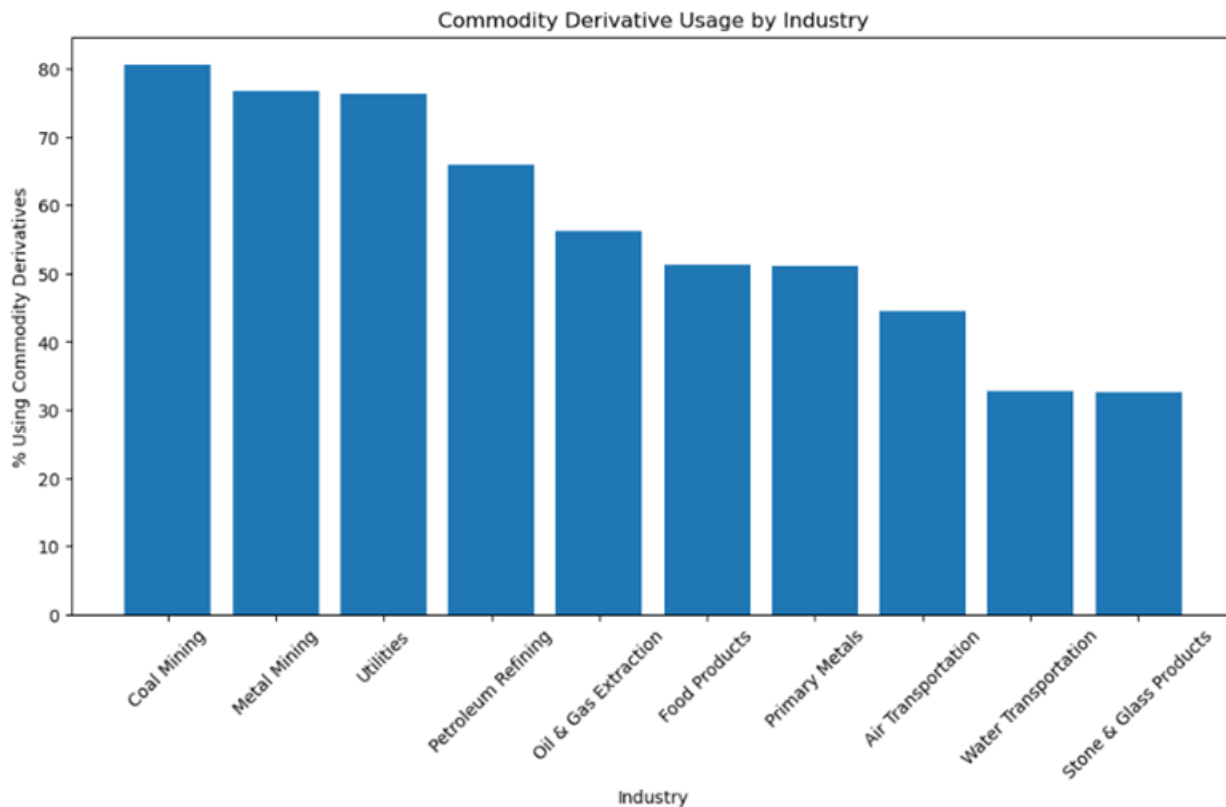


Figure 11 shows the percentage of firms using commodity derivatives across major 2-digit SIC industries. Commodity derivative usage appears highest in industries with significant exposure to commodity price risk, including coal mining, metal mining, utilities, and petroleum refining. The relatively high adoption rates observed in these industries suggest that commodity derivatives play an important role in managing input cost uncertainty and commodity price volatility.

The results are consistent with the characteristics of industries that are more directly affected by changes in commodity prices and raw material costs. Firms operating in energy, mining, transportation, and materials-related industries generally face greater exposure to commodity market fluctuations and therefore have stronger incentives to hedge these risks. In contrast, industries with less direct exposure to commodity price movements reported lower rates of derivative usage. Overall, the findings indicate that commodity derivative usage is concentrated among firms operating in industries with greater sensitivity to commodity market volatility.

Figure 12: Commodity Derivative Use by Notional Amount

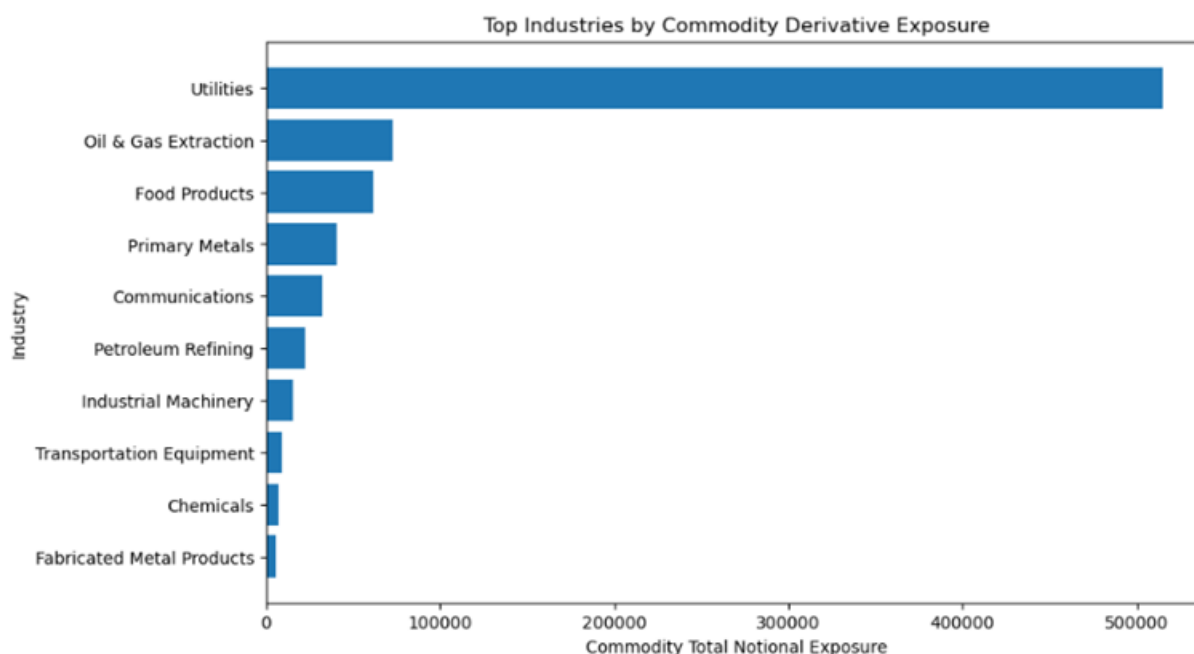


Figure 12 reports the industries with the highest aggregate commodity derivative notional exposure across the full sample period. Utilities reported the largest cumulative commodity derivative positions, followed by Oil & Gas Extraction, Food Products, Primary Metals, and Communications. The concentration of exposure within these industries suggests that commodity derivative usage is closely related to firms' exposure to energy, fuel, raw materials, and other commodity related inputs.

While several industries reported high rates of commodity derivative usage, aggregate exposure was heavily concentrated among a smaller group of firms maintaining large notional positions. Utilities, in particular, accounted for substantially larger notional exposure than other industries, reflecting the importance of managing fuel and energy price volatility.

The results also reveal an important distinction between derivative adoption and derivative exposure. While Coal Mining and Metal Mining reported some of the highest rates of commodity derivative usage, Utilities and Oil & Gas Extraction maintained the largest aggregate notional exposures. This suggests that industries with lower adoption rates may still hold significantly larger derivative positions when commodity derivatives are utilized.

Regression Analysis

Table 6: Regression Analysis of Commodity Derivative Exposure

Variable	Coefficient	t-Statistic	p-value
Intercept	-103.3872	-3.2068	0.0014
Year	0.0535	3.3388	0.0009

Table 6 presents the regression results examining the relationship between time and commodity derivative notional amounts. The dependent variable is the logarithm of positive commodity derivative notional amounts. The coefficient on Year is positive and statistically significant at the 1% level, indicating that commodity derivative positions generally increased over the sample period. The estimated coefficient suggests that

commodity derivative notional amounts increased by approximately 5.35% per sample year among firms reporting commodity derivative usage. Although the model has a relatively low R-squared value of 0.025, this is not unexpected given the substantial variation in hedging activity across firms and industries. The regression results indicate a positive relationship between time and commodity derivative exposure among firms reporting commodity derivative usage.

Overall Comparison of Derivative Usage

This section compares the usage patterns of interest rate, currency, and commodity derivatives among non-financial S&P 1500 firms over the sample period. While all three derivative types are used for risk management purposes, substantial differences exist in both adoption rates and notional exposure levels across derivative categories.

Interest rate and currency derivatives were consistently used by a larger number of firms than commodity derivatives throughout the sample period. Interest rate derivatives exhibited the highest levels of aggregate notional exposure in most years, reflecting the importance of managing financing costs and debt-related risks. Currency derivatives also maintained relatively high adoption rates and experienced substantial growth in notional exposure, particularly after 2012, as firms became increasingly exposed to global markets and foreign exchange risk.

Commodity derivative usage remained comparatively limited and was concentrated among firms operating in industries directly affected by commodity price fluctuations. Despite lower adoption rates, firms using commodity derivatives often maintained substantial derivative positions. Industry analyses showed that commodity derivative activity was concentrated in sectors such as utilities, oil and gas extraction, and petroleum refining, where firms face greater exposure to energy, fuel, and raw material price volatility.

Overall, the findings suggest that derivative usage varies according to the nature of the risks firms face. Interest rate and currency derivatives appear to address broader risk management needs across a wide range of industries, while commodity derivatives are more specialized and concentrated among firms with significant exposure to commodity markets.

Firm Clustering Analysis

To further examine similarities and differences in derivative usage behavior, firms were grouped using clustering techniques based on their derivative exposures. The clustering analysis identifies distinct groups of firms characterized by different levels and combinations of interest rate, currency, and commodity derivative usage.

Table 7: Clustering Between Users

Cluster Type	Avg Interest Rate Exposure	Avg Currency Exposure	Avg Commodity Exposure
Primarily Currency Users	1.72	848.25	0.28
Interest Rate & Currency Users	1806.74	1050.34	0.17
Multi-Derivative Users	1438.54	1413.15	2481.83

Table 7 presents the average derivative exposures for the three clusters identified using K-means clustering. The first cluster, labeled Primarily Currency Users, consists of firms with substantial currency derivative exposure but relatively limited interest rate and commodity derivative activity. The second cluster, Interest Rate & Currency Users, represents the largest group of firms and is characterized by significant exposure to both interest rate and currency derivatives. The third cluster, Multi-Derivative Users, contains firms with large exposures across all three derivative categories and exhibits the highest average commodity derivative exposure.

The clustering results suggest that derivative usage among non-financial firms is not uniform. Most firms are concentrated in clusters focused on interest rate and currency risk management, while only a small subset of firms maintain significant exposure across all three derivative categories. These findings are consistent with the earlier analysis, which showed that commodity derivatives are less widely used than interest rate and currency derivatives and tend to be concentrated among firms with more extensive risk management programs.

Figure 13: Derivative Users Clustering

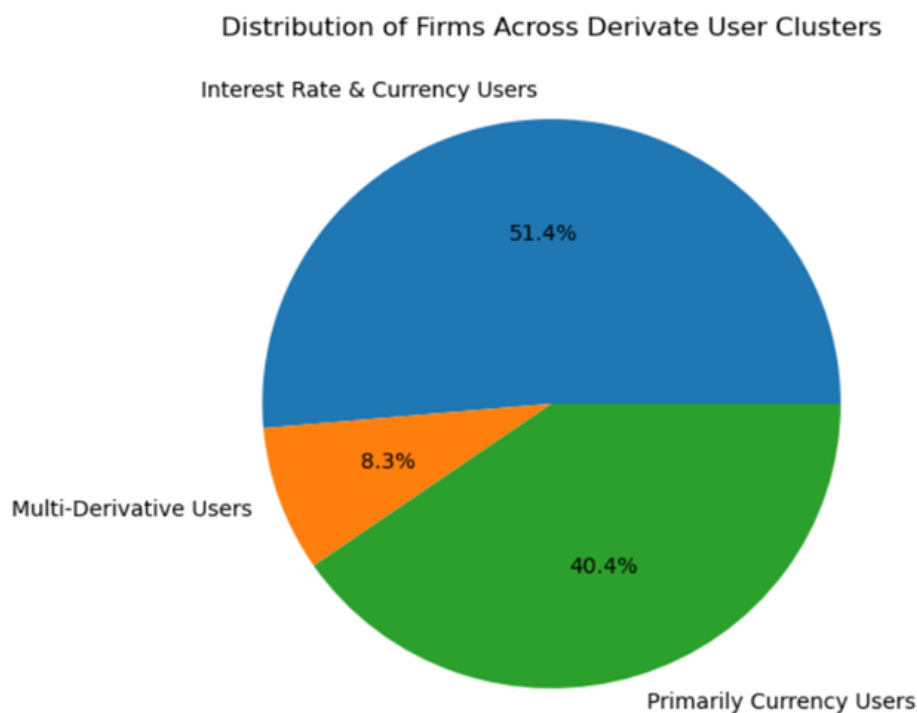


Figure 13 presents the distribution of firms across the three clusters identified using K-means clustering. Approximately 51.4% of firms were classified as Interest Rate & Currency Users, while 40.4% were classified as Primarily Currency Users. Only 8.3% of firms were categorized as Multi-Derivative Users, indicating that extensive use of interest rate, currency, and commodity derivatives simultaneously is relatively uncommon. Overall, the results suggest that most firms focus their risk management activities on interest rate and currency exposures, while a smaller group of firms maintain broader derivative programs spanning multiple sources of risk.

Conclusion

This analysis examined the use of interest rate, currency, and commodity derivatives among non-financial S&P 1500 firms from 2000 through 2024. Using time-series, cross-sectional, industry, regression, and clustering analyses, the results demonstrate that derivative usage varies considerably across derivative types, industries, and firms. While all three derivative categories serve important risk management functions, firms appear to utilize these instruments differently depending on their exposure to financial and operational risks.

Interest rate and currency derivatives were consistently the most widely used instruments and accounted for the largest aggregate notional exposures throughout the sample period. Commodity derivatives were used by a smaller percentage of firms and were concentrated within industries facing greater exposure to commodity price volatility, including utilities, oil and gas extraction, and petroleum refining. Despite lower adoption rates, firms using commodity derivatives often maintained substantial derivative positions, highlighting the importance of commodity risk management for certain sectors.

The results further indicate that derivative usage is highly heterogeneous across firms. Clustering analysis revealed that most firms were concentrated in groups focused primarily on interest rate and currency risk management, while only a small subset of firms maintained significant exposure across all three derivative categories. These findings suggest that extensive derivative usage tends to be concentrated among firms facing more complex combinations of financial and operational risks.

Overall, the findings demonstrate that corporate derivatives remain an important tool for managing uncertainty and reducing exposure to interest rate, currency, and commodity price fluctuations. As financial markets continue to evolve and firms face increasing exposure to global economic risks, derivative usage is likely to remain an important component of corporate risk management strategies. Future research could examine how derivative usage changes in response to major economic events, interest rate cycles, geopolitical shocks, and evolving regulatory environments.